

MATHEMATICS

TARGET : JEE- Advanced

CAPS-9**BINOMIAL THEOREM****SCQ (Single Correct Type) :**

- If a, b, c, d be four consecutive coefficients in the binomial expansion of $(1+x)^n$, then value of the expression $\left(\left(\frac{b}{b+c}\right)^2 - \frac{ac}{(a+b)(c+d)}\right)$ (where $x > 0$ and $n \in \mathbb{N}$) is
 (A) Positive (B) negative (C) zero (D) depend on n
- The coefficient of x^{28} in the expansion of $(2 - x^3 + x^6)^{30}$ is
 (A) 0 (B) 1 (C) 14 (D) 28
- If each coefficient in the expansion of the expression $x(1+x)^n$ ($n \in \mathbb{N}$) in powers of x is divided by the exponent of corresponding power, then the sum of the values thus obtained is equal to _____.
 (A) $\frac{2^n}{n+1}$ (B) $\frac{2^n - 1}{n+1}$ (C) $\frac{2^n + 1}{n+1}$ (D) $\frac{2^{n+1} - 1}{n+1}$
- The number of distinct terms in the expansion of $(x + y^2)^{13} + (x^2 + y)^{14}$ is _____.
 (A) 27 (B) 29 (C) 28 (D) 25
- If $f(x) = \sum_{r=1}^n \left[r \left({}^{n-1}C_{r-1} - r \cdot {}^nC_{r-1} \right) + (2r+1) {}^nC_r \right]$ then _____.
 (A) $f(n) = n^2 - 1$ (B) $f(n) = (n+1)^2 - 1$
 (C) $f(n) = (n+1)^2 + 1$ (D) $\sum_{n=1}^{10} f(n) = 374$
- If $(1+px+x^2)^n = 1 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, where $n \in \mathbb{N}$, $p \in \mathbb{R}$. If $p = -3$ and n is even number, then the value of $a_1 + 3a_2 + 5a_3 + 7a_4 + \dots + (4n-1)a_{2n}$ is _____.
 (A) n (B) $2n - 1$ (C) $2n - 2$ (D) $2n$
- If $6^{83} + 8^{83}$ is divided by 49, then the remainder is
 (A) 35 (B) 5 (C) 1 (D) 0
- The sum of the series $(1^2 + 1).1! + (2^2 + 1).2! + (3^2 + 1).3! + \dots + (n^2 + 1).n!$
 (A) $(n+1).(n+2)!$ (B) $n.(n+1)!$ (C) $(n+1).(n+1)!$ (D) none of these

MCQ (One or more than one correct) :

9. If $J_m = \sum_{r=0}^{m-3} {}^m C_r {}^m C_{r+3}$ and m_1, m_2 are two values of m satisfying $5 J_m = 3 J_{m+1}$, then correct statement is/are (where $[.]$ denotes greatest integer function)
- (A) $m_1 + m_2 = \frac{-8}{7}$ (B) $m_1 m_2 = \frac{46}{7}$
- (C) $[(1-m_1)(1-m_2)] = 8$ (D) $[(1+m_1)(1+m_2)] = 8$
10. For natural number m, n if $(1-y)^m (1+y)^n = 1 + a_1 y + a_2 y^2 + \dots$ and $a_1 = a_2 = 10$ then
- (A) $m < n$ (B) $m > n$ (C) $m + n = 80$ (D) $m - n = 20$
11. Which of the following is/are true
- (A) $5^6 - {}^5 C_1 \cdot 4^6 + {}^5 C_2 \cdot 3^6 - {}^5 C_3 \cdot 2^6 + {}^5 C_4 \cdot 1^6 = {}^6 C_2 \cdot 5$
- (B) $6^5 - {}^6 C_1 \cdot 5^5 + {}^6 C_2 \cdot 4^5 - {}^6 C_3 \cdot 3^5 + {}^6 C_4 \cdot 2^5 - {}^6 C_5 \cdot 1^5 = 0$
- (C) $6^6 - {}^6 C_1 \cdot 5^6 + {}^6 C_2 \cdot 4^6 - {}^6 C_3 \cdot 3^6 + {}^6 C_4 \cdot 2^6 - {}^6 C_5 \cdot 1^6 = 720$
- (D) $6^5 - {}^6 C_1 \cdot 5^5 + {}^6 C_2 \cdot 4^5 - {}^6 C_3 \cdot 3^5 + {}^6 C_4 \cdot 2^5 - {}^6 C_5 \cdot 1^5 = {}^5 C_2 \cdot 6$
12. In the expansion of $\left(x^3 + 3 \cdot 2^{-\log_{\sqrt{2}} \sqrt{x^3}}\right)^{11}$
- (A) there appears a term with the power x^2
- (B) there does not appear a term with the power x^2
- (C) there appears a term with the power x^{-3}
- (D) the ratio of the co-efficient of x^3 to that of x^{-3} is $1/3$
13. If $(9 + \sqrt{80})^n = I + f$ where I, n are integers and $0 < f < 1$, then
- (A) I is an odd integer (B) I is an even integer
- (C) $(I + f)(1 - f) = 1$ (D) $1 - f = (9 - \sqrt{80})^n$

Comprehension Type Question:**Comprehension # 1**

If $(1 + x + x^2)^{2n} = 1 + a_1 x + a_2 x^2 + \dots + a_{4n} x^{4n}$, then

14. The value of $\sum_{r=0}^{n-1} a_{2r}$, is
- (A) $\frac{9^n + 1 - 2a_{2n}}{4}$ (B) $\frac{9^n + 1 + 2a_{2n}}{4}$ (C) $\frac{9^n + 1 - 2a_n}{4}$ (D) $\frac{9^n - 1 - 2a_{2n}}{4}$
15. What value a_2 takes ?
- (A) $2^n C_2$ (B) $2^{n+1} C_2$ (C) $2^{n-1} C_2$ (D) $n C_2$

Numerical based Questions :

16. If the coefficient of x^5 in $(1+x)^{21} + (1+x)^{22} + \dots + (1+x)^{30}$ is $\lambda {}^nC_r + \mu {}^mC_r$, λ, μ, n, r, m being integers and ${}^nC_r, {}^mC_r$ are binomial coefficients, then the value of $\lambda(n+r) + \mu(m+r)$ is
17. The value of $\sum_{r=0}^9 \frac{{}^{10}C_r}{{}^{10}C_r + {}^{10}C_{r+1}}$ is equal to
18. When the terms in the binomial expansion of $\left(\sqrt{x} + \frac{1}{2x^{1/4}}\right)^n$ are arranged in decreasing powers of x , the coefficients of the first three terms are in arithmetic progression. The number of terms in the expansion with integer powers of x is _____.
19. Find the coefficient of x^7 in $(1-x+2x^3)^{10}$.
20. Find the coefficient of x^{49} in the polynomial

$$\left(x - \frac{C_1}{C_0}\right) \left(x - 2^2 \cdot \frac{C_2}{C_1}\right) \left(x - 3^2 \cdot \frac{C_3}{C_2}\right) \dots \dots \dots \left(x - 50^2 \cdot \frac{C_{50}}{C_{49}}\right) \text{ where } C_r = 50C_{r-1}.$$

Matrix Match Type :

21. Match the following:

Column – I

(A) Number of distinct terms in the expansion of $(x+y-z)^{16}$ is

(B) Number of terms in the expansion of

$$\left(x + \sqrt{x^2 - 1}\right)^6 + \left(x - \sqrt{x^2 - 1}\right)^6 \text{ is}$$

(C) The number of irrational terms in $\left(\sqrt[8]{5} + \sqrt[6]{2}\right)^{100}$ is

(D) The sum of numerical coefficients in the expansion of

$$\left(1 + \frac{x}{3} + \frac{2y}{3}\right)^{12} \text{ is}$$

Column – II

(p) 2^{12}

(q) 97

(r) 4

(s) 153

Options:

- (A) A $\rightarrow$ (s); B $\rightarrow$ (r); C $\rightarrow$ (q); D $\rightarrow$ (p)
 (B) A $\rightarrow$ (s); B $\rightarrow$ (r); C $\rightarrow$ (p); D $\rightarrow$ (q)
 (C) A $\rightarrow$ (r); B $\rightarrow$ (s); C $\rightarrow$ (q); D $\rightarrow$ (p)
 (D) A $\rightarrow$ (s); B $\rightarrow$ (q); C $\rightarrow$ (r); D $\rightarrow$ (p)

22. Match the following:

Column – I

Column – II

- (A) The minimum value of $a + b + c + d$ if $\log_3(a + b) + \log_3(c + d) \geq 4$ is (p) 18
- (B) The number of distinct terms in the expansion of $\left(x + y + \frac{1}{x} + \frac{1}{y}\right)^{14}$ is (q) 225
- (C) The remainder when $(23)^{86}$ is divided by 100 is (r) 89
- (D) If the third term in the expansion of $\left(\frac{1}{x} + x^{\log_{10} x}\right)^5$ is 1000 and $x > 1$, then the value of x is (s) 100
- (A) $A \rightarrow (p); B \rightarrow (q); C \rightarrow (s); D \rightarrow (r)$ (B) $A \rightarrow (p); B \rightarrow (r); C \rightarrow (q); D \rightarrow (s)$
- (C) $A \rightarrow (p); B \rightarrow (q); C \rightarrow (r); D \rightarrow (s)$ (D) $A \rightarrow (q); B \rightarrow (p); C \rightarrow (r); D \rightarrow (s)$

Subjective based Questions:

23. (a) Find the index of n of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the 9th term of the expansion has numerically greatest coefficient ($n \in \mathbb{N}$).
- (b) For what positive value of x is the fourth term in the expansion of $(5 + 3x)^9$ is the greatest.
24. Show that $(1 \cdot 2)^n C_2 + (2 \cdot 3)^n C_3 + \dots + (n - 1) \cdot n^n C_n = n(n - 1)2^{n-2}$
25. If $a_1, a_2, a_3, \dots, a_n$ are in A.P. and S_n is the sum of first n term's, show that
- $$\sum_{k=0}^n {}^n C_k S_k = 2^{n-2} (na_1 + S_n)$$
26. Show that $S_1 = \sum_{r=1}^{2n} {}^{4n} C_{2r-1} (-1)^{r-1} = 0$
27. In the expansion of $(x + y)^n$, if the sum of odd term's be p and sum of even term's be q , then prove that
- (i) $p^2 - q^2 = (x^2 - y^2)^n$ (ii) $4pq = (x + y)^{2n} - (x - y)^{2n}$

ANSWER KEY OF CAPS-9

- | | | | | |
|--------------|-----------|-----------|--|----------|
| 1. (A) | 2. (A) | 3. (D) | 4. (C) | 5. (B) |
| 6. (D) | 7. (A) | 8. (B) | 9. (ABC) | 10. (AC) |
| 11. (ABC) | 12. (BCD) | 13. (ACD) | 14. (A) | 15. (B) |
| 16. (10) | 17. (5) | 18. (3) | 19. (960) | |
| 20. (-22100) | 21. (A) | 22. (C) | 23. (a) $n=12$, (b) $\left\{\frac{5}{7} \leq x \leq \frac{10}{9}\right\}$ | |